

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT

On

ANALYSIS AND DESIGN OF ALGORITHMS (23CS4PCADA)

Submitted by

KRATI GOEL (1BF24CS148)

**in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING**



**B.M.S. COLLEGE OF ENGINEERING
(Autonomous Institution under VTU)
BENGALURU-560019
February to June 2026**

CERTIFICATE

**B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
(Affiliated To Visvesvaraya Technological University, Belgaum)
Department of Computer Science and Engineering**



This is to certify that the Lab work entitled “**ANALYSIS AND DESIGN OF ALGORITHMS**” carried out by Krati Goel (**1BF24CS148**), who is a bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2025-26. The Lab report has been approved as it satisfies the academic requirements in respect of Analysis and Design of Algorithms Lab - (**23CS4PCADA**) work prescribed for the said degree.

Dr. Manjula S
Assistant Professor
Department of CSE
BMSCE, Bengaluru

Dr. Kavitha Sooda
Professor and Head
Department of CSE
BMSCE, Bengaluru

Index Sheet

Sl. No.	Experiment Title	Page No.
1	Write a program to obtain the Topological ordering of vertices in a given digraph.	4-7
2	Implement Johnson Trotter algorithm to generate permutations.	8-9
3	Sort a given set of N integer elements using Merge Sort technique and compute its time taken. Run the program for different values of N and record the time taken to sort.	10-13
4	Sort a given set of N integer elements using Quick Sort technique and compute its time taken.	14-17
5	Sort a given set of N integer elements using Heap Sort technique and compute its time taken.	18-20
6	Implement 0/1 Knapsack problem using dynamic programming. (5 Marks)	21-22
7	Implement All Pair Shortest paths problem using Floyd's algorithm.	23-26
8	(a) Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm. (b) Find Minimum Cost Spanning Tree of a given undirected graph using Kruskal's algorithm.	27-28 28-30
9	Implement Fractional Knapsack using Greedy technique.	31-34
10	From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.	35-36
11	Implement "N-Queens Problem" using Backtracking.	37-38

Course outcomes:

CO1	Analyze time complexity of algorithms using asymptotic notations.
CO2	Apply various algorithm design techniques for the given problem.
CO3	Evaluate the appropriate algorithm/design technique for solving a given problem.
CO4	Conduct practical experiments by applying appropriate algorithm design technique to solve problems.

Lab program 1:

Topological Sorting

```
#include <stdio.h>
#define MAX 10
int graph[MAX][MAX], indegree[MAX], queue[MAX];
int n;
void topologicalSort()
{
    int front = 0, rear = 0;
    int count = 0, i, j;
    // Find indegree of each vertex
    for (i = 0; i < n; i++) {
        indegree[i] = 0;
        for (j = 0; j < n; j++)
            indegree[i] += graph[j][i];
    }
    // Insert all vertices with indegree 0 into queue
    for (i = 0; i < n; i++)
        if (indegree[i] == 0)
            queue[rear++] = i;
    printf("Topological Ordering: ");
    while (front < rear) {
        int v = queue[front++];
        printf("%d ", v);
        count++;
        // Reduce indegree of adjacent vertices
        for (i = 0; i < n; i++)
            if (graph[v][i] == 1) {
                indegree[i]--;
                if (indegree[i] == 0)
                    queue[rear++] = i;
            }
    }
}
```

```

        // Check for cycle
        if (count != n)
            printf("\nGraph contains a cycle. Topological sort not possible.\n");
    }

int main()
{
    int i, j;

    printf("Enter number of vertices: ");
    scanf("%d", &n);
    printf("Enter adjacency matrix:\n");
    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &graph[i][j]);
    topologicalSort();

    return 0;
}

```

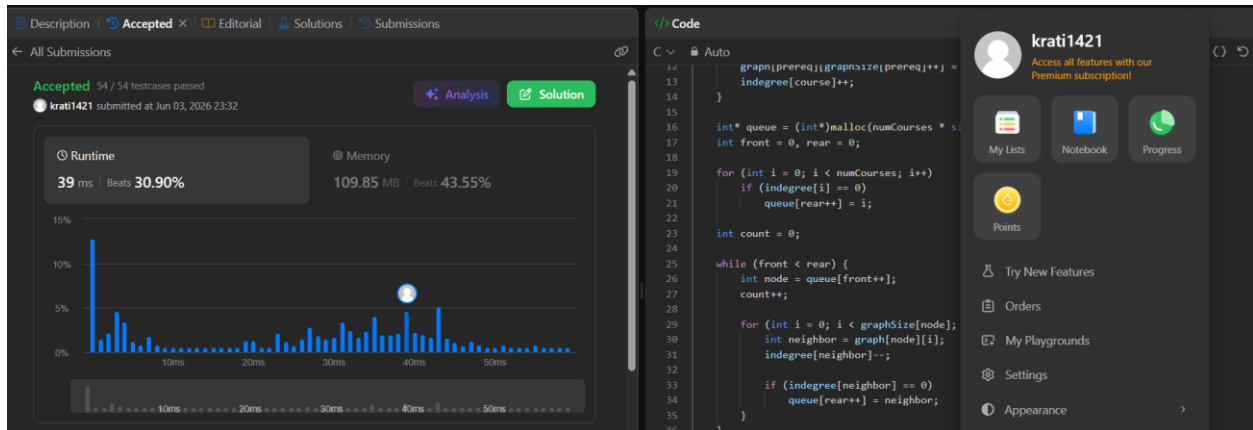
OUTPUT:

```

Enter number of vertices: 6
Enter adjacency matrix:
0 1 1 0 0 0
0 0 0 1 1 0
0 0 0 1 0 0
0 0 0 0 0 1
0 0 0 0 0 1
0 0 0 0 0 0
Topological Ordering: 0 1 2 4 3 5

```

LEETCODE 207



```
bool canFinish(int numCourses, int** prerequisites, int prerequisitesSize, int*
prerequisitesColSize) {
```

```
    int** graph = (int**)malloc(numCourses * sizeof(int*));
```

```
    int* indegree = (int*)calloc(numCourses, sizeof(int));
```

```
    int* graphSize = (int*)calloc(numCourses, sizeof(int));
```

```
    for (int i = 0; i < numCourses; i++)
```

```
        graph[i] = (int*)malloc(numCourses * sizeof(int));
```

```
    for (int i = 0; i < prerequisitesSize; i++)
```

```
        { int course = prerequisites[i][0];
```

```
          int prereq = prerequisites[i][1];
```

```
          graph[prereq][graphSize[prereq]++] = course;
```

```
          indegree[course]++;
```

```
    }
```

```
    int* queue = (int*)malloc(numCourses * sizeof(int));
```

```
    int front = 0, rear = 0;
```

```
    for (int i = 0; i < numCourses; i++)
```

```
        if (indegree[i] == 0)
```

```
            queue[rear++] = i;
```

```
int count = 0;
while (front < rear) {
    int node = queue[front++];
    count++;

    for (int i = 0; i < graphSize[node]; i++)
        { int neighbor = graph[node][i];
          indegree[neighbor]--;
          if (indegree[neighbor] == 0)
              queue[rear++] = neighbor;
          }
    }
return count == numCourses;
}
```

Lab program 2:

Johnson Trotter

```
#include <stdio.h>
```

```
// -1 means left
```

```
int mobile(int p[], int d[], int
n){ int m = 0;
for (int i = 0; i < n; i++) {
    if (d[i] == -1 && i > 0 && p[i] > p[i - 1] && p[i] > m)
        m = p[i];
    if (d[i] == 1 && i < n - 1 && p[i] > p[i + 1] && p[i] > m)
        m = p[i];
}
return m;
}
```

```
int
main(){
    int n;
    int p[10], d[10];
    int i, m, pos, next, temp;
    printf("Enter n:");
    scanf("%d", &n);
    for (i = 0; i < n;
        i++){ p[i] = i + 1;
        d[i] = -1;
    }

    while (1)
    {
        for (i = 0; i < n;
            i++){ printf("%d ", p[i]);
        }

        printf("\n");
        m = mobile(p, d, n);
```



```

    if (m == 0)
        break;
    for (i = 0; i < n; i++)
        if (p[i] == m)
            pos = i;
    next = pos + d[pos];
    temp = p[pos];
    p[pos] = p[next];
    p[next] = temp;

    temp = d[pos];
    d[pos] = d[next];
    d[next] = temp;

    for (i = 0; i < n; i++)
        if (p[i] > m)
            d[i] = -d[i];
}
return 0;
}

```

OUTPUT

```

Enter n:3
1 2 3
1 3 2
3 1 2
3 2 1
2 3 1
2 1 3

```

Lab Program 3:

Merge Sort

```
#include<stdlib.h>
```

```
#include <stdio.h>
```

```
#include <time.h>
```

```
void merge (int low,int mid,int high, int
```

```
    *a){ int i=low, j=mid+1,k=low;
```

```
    int arr[high+1];
```

```
    while(i<=mid && j<= high){
```

```
        if(a[i]< a[j]) arr[k++]=a[i++];
```

```
        else    arr[k++]=a[j++];
```

```
    }
```

```
    while(i<=mid) arr[k++]=a[i++];
```

```
    while(j<=high) arr[k++]=a[j++];
```

```
    for(i=low;i<=high;i++) a[i]=arr[i];
```

```
}
```

```
void mergeSort(int low,int high, int
```

```
    *a){ if(low == high) return;
```

```
    int mid =(low + high)/2;
```

```
    mergeSort(low,mid,a);
```

```
    mergeSort(mid+1,high,a);
```

```
    merge(low, mid, high, a);
```

```
}
```

```
int
```

```
    main(){
```

```
        int n;
```

```
        printf("Enter size :");
```

```
        scanf("%d",&n);
```

```
        int a[n];
```

```

// printf("Enter %d items:\n",n);
for(int i=0; i<n;i++)
    a[i]=rand();
    // scanf("%d",&a[i]);

double stTime =clock();
mergeSort(0,n-1,a);
double endTime =clock();
double t= (endTime -stTime)/CLOCKS_PER_SEC;
// printf("Sorted Array :\n");
// for(int i=0;i<n;i++)
//     printf("%d\t",a[i]);
printf("\nTime Taken: %.4f ms",t*1000);
return 0;
}

```

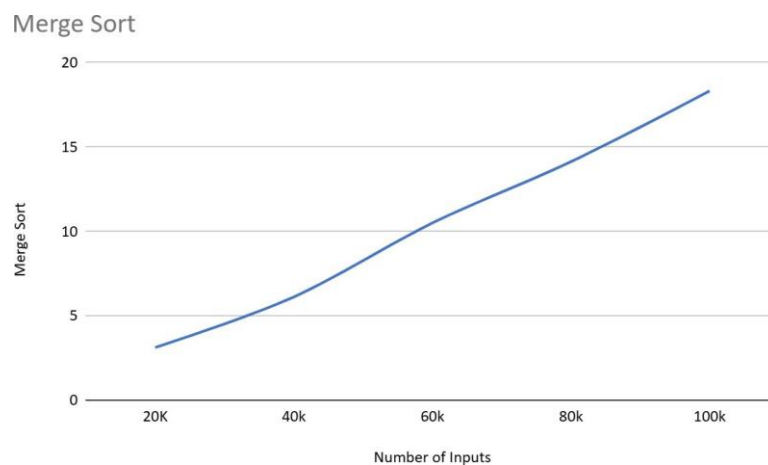
OUTPUT:

```

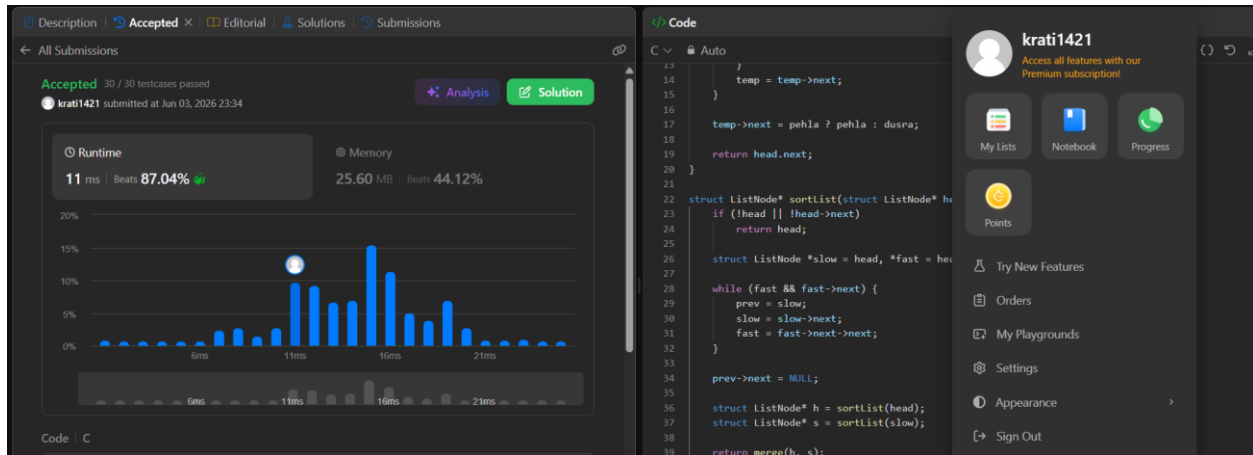
Enter size :5
Enter 5 items:
20 25 15 10 17
Sorted Array :
10 15 17 20 25
Time Taken: 0.0020 ms

```

GRAPH:



LEETCODE 148



/**

* Definition for singly-linked list.

* struct ListNode {

* int val;

* struct ListNode *next;

* };

*/

```
struct ListNode* merge(struct ListNode * pehla , struct ListNode *
dusra){ struct ListNode head;
struct ListNode * temp = &head ;
head.next = NULL ;
```

```
while(pehla &&
dusra){ if(pehla->val <
dusra->val){
temp->next = pehla;
pehla = pehla->next;
}
else{
temp->next =dusra;
dusra = dusra->next;
```

```

    }
    temp = temp->next;
}

temp->next= pehla ? pehla : dusra ;

return head.next;
}

struct ListNode* sortList(struct ListNode* head)
{ if(!head || !head->next) return head;

    struct ListNode * slow =head , *fast = head , * prev = NULL;
    while(fast && fast->next){
        prev = slow;
        slow = slow->next;
        fast = fast->next->next ;
    }

    prev-> next=NULL;
    struct ListNode* h = sortList(head);
    struct ListNode* s = sortList(slow);
    return merge(h,s);
}

```

Lab program 4:

Quick Sort

```
#include<stdlib.h>
```

```
#include <stdio.h>
```

```
#include <time.h>
```

```
void swap(int a[], int i,int  
    j){ int t=a[i];  
    a[i]=a[j];  
    a[j]=t;  
}
```

```
int sort(int a[],int st,int  
    end){ int  
    p=a[st],i=st,j=end; while  
    (1) {  
        while (i <= end && a[i] <= p) i++;  
        while (j >= st && a[j] > p) j--;  
  
        if (i >= j) break;  
        swap(a, i, j);  
    }  
    swap(a,st,j);  
    return j;  
}
```

```
void quickSort(int a[],int st,int  
    end ){ if(st >= end) return;  
  
    int mid = sort(a,st,end);  
    quickSort(a,st,mid-1);  
    quickSort(a,mid+1,end);  
}
```

```
int main(){
```

```

int n;
printf("Enter size :");
scanf("%d",&n);
int a[n];
// printf("Enter %d items :\n", n);
for(int i=0; i<n;i++)
    a[i]=rand();
    // scanf("%d",&a[i]);

double stTime =clock();
quickSort(a,0,n-1);
double endTime =clock();
double t= (endTime -stTime)/CLOCKS_PER_SEC;
// printf("Sorted Array :\n");
// for(int i=0;i<n;i++)
    // printf("%d\t",a[i]);
printf("\nTime Taken: %.4f ms",t*1000);
return 0;
}

```

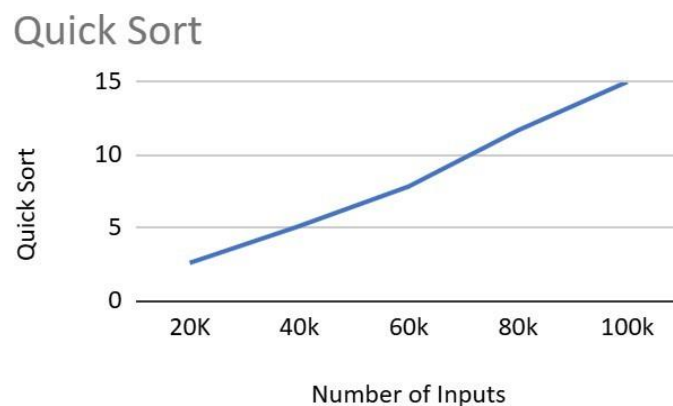
OUTPUT:

```

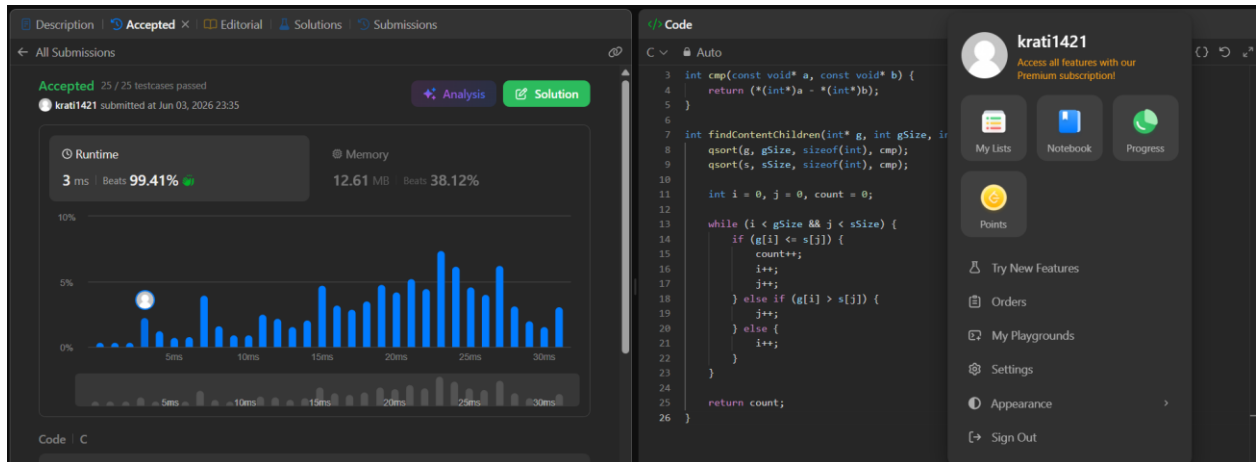
Enter size :5
Enter 5 items :
80 16 17 19 18
Sorted Array :
16 17 18 19 80
Time Taken: 0.0030 ms

```

GRAPH:



LEETCODE 455



```
void swap(int a[], int i, int
j){ int t=a[i];
    a[i]=a[j];
    a[j]=t;
}
```

```
int sort(int a[],int st,int
end){ int
p=a[st],i=st,j=end; while
(1) {
    while (i <= end && a[i] <= p) i++;
    while (j >= st && a[j] > p) j--;

    if (i >= j) break;
    swap(a, i, j);
}
swap(a,st,j);
return j;
}
```

```
void quickSort(int a[],int st,int end ){
```



```

    if(st >= end) return;

    int mid = sort(a,st,end);
    quickSort(a,st,mid-1);
    quickSort(a,mid+1,end);
}

int findContentChildren(int* g, int gSize, int* s, int sSize)
{
    quickSort(g, 0, gSize-1);
    quickSort(s, 0, sSize-1);
    int count=0, i=0, j=0;
    while(i<gSize &&
        j<sSize){ if(g[i]<=s[j]) {
        count++;i++;j++;
        }
        else j++;
    }
    return count;
}

```

Lab program 5:

Heap Sort

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <time.h>
```

```
void heapify(int arr[], int n, int i)
```

```
{ int lar = i;
```

```
int left = 2 * i + 1;
```

```
int right = 2 * i + 2;
```

```
if (left < n && arr[left] > arr[lar])
```

```
lar = left;
```

```
if (right < n && arr[right] > arr[lar])
```

```
lar = right;
```

```
if (lar != i)
```

```
{ int t =
```

```
arr[i];
```

```
arr[i] = arr[lar];
```

```
arr[lar] = t;
```

```
heapify(arr, n, lar);
```

```
}
```

```
}
```

```
void heapSort(int arr[], int n)
```

```
{ for (int i = n / 2 - 1; i >= 0; i--)
```

```
-)
```

```
heapify(arr, n, i);
```

```
for (int i = n - 1; i > 0; i--)
```

```
{ int t = arr[0];
```

```
arr[0] = arr[i];
```

```
arr[i] = t;
```

```
heapify(arr, i, 0);
```

```
}
```

```
}
```

```

int main()
{
    int n;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Invalid number of elements.\n");
        return 1;
    }
    int *arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }
    // printf("Enter %d integer elements:\n", n);
    for (int i = 0; i < n; i++)
        arr[i]=rand();
    // scanf("%d", &arr[i]);
    clock_t start = clock();
    heapSort(arr, n);
    clock_t end = clock();
    double time_taken = (double)(end - start) / CLOCKS_PER_SEC;
    // printf("\nSorted elements:\n");
    // for (int i = 0; i < n; i++)
    //     printf("%d ", arr[i]);
    printf("\n\nTime taken: %.3f ms\n", time_taken*1000);
    return 0;
}

```

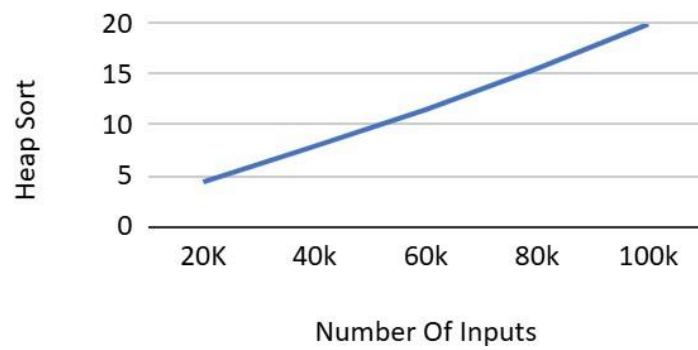
OUTPUT:

```
Enter number of elements: 7
Enter 7 integer elements:
15 10 20 17 8 25 5
|
Sorted elements:
5 8 10 15 17 20 25

Time taken: 0.002 ms
```

GRAPH:

Heap Sort



Lab program 6:

Knapsack 0/1 (using DP)

```
#include <stdio.h>

int max(int a , int
    b){ return
    (a>b)?a:b;
}

int knapsack(int W, int wt[], int val[], int
    n){ int dp[n+1][W+1];
    for(int i=0;i<=n;i++)
        for(int w=0;w<=W;w++)
            if(i==0 || w==0) dp[i][w]=0;
            else if(wt[i-1]<=w) dp[i][w]=max(val[i-1]+dp[i-1][w-wt[i-1]], dp[i-1][w]);
            else dp[i][w]= dp[i-1][w];
    for(int
        i=0;i<=n;i++){ for(int
        w=0;w<=W;w++)
            printf("%d\t",dp[i][w]);
            printf("\n");
        }
    return dp[n][W];
}

void
    main(){ int
    n, W;
    printf("Enter number of items:");
    scanf("%d",&n);
    printf("Enter max capacity:");
    scanf("%d",&W);
    int val[n];
    int wt[n];
    printf("Enter wt and value for %d items:\n",n);
    for(int i=0;i<n;i++)
        scanf("%d %d",&wt[i], &val[i]);
```

```
    printf("Result = %d", knapsack(W, wt, val ,n));  
}
```

OUTPUT:

```
Enter number of items:4
Enter max capacity:5
Enter wt and value for 4 items:
2 8
1 6
3 16
2 11
0 0 0 0 0 0
0 0 8 8 8 8
0 6 8 14 14 14
0 6 8 16 22 24
0 6 11 17 22 27
Result = 27
```

LEETCODE 70

The screenshot shows the LeetCode interface for problem 70, "Climbing Stairs". The submission is marked as "Accepted" with 45/45 test cases passed. The runtime is 0 ms, which beats 100.00% of other submissions. The memory usage is 8.60 MB, which beats 36.78%. A bar chart shows the runtime performance relative to other submissions. The code editor displays a C++ solution using dynamic programming. The user's profile "krati1421" is visible on the right.

```
1 int climbStairs(int n) {
2     if (n <= 2)
3         return n;
4     int dp[n + 1];
5
6     dp[1] = 1;
7     dp[2] = 2;
8
9     for (int i = 3; i <= n; i++)
10         dp[i] = dp[i - 1] + dp[i - 2];
11
12     return dp[n];
13 }
```

```
int climbStairs(int n)
{ if (n <= 2) return
n; int dp[n + 1];
dp[1] = 1;dp[2] = 2;
for (int i = 3; i <= n; i++) dp[i] = dp[i - 1] + dp[i - 2];
return dp[n];
}
```

Lab program 7:

Floyd's Algorithm

```
#include <stdio.h>

#define N 10

#define INFI 99999

int min(int a, int
b){ return (a<b)? a:
        b;
}

void floyd(int d[N][N], int
n){ for (int k=0;k<n;k++)
    for(int i=0;i<n;i++)
        for(int j=0;j<n;j++)
            d[i][j] = min(d[i][j], d[i][k] + d[k][j]);

    printf("\nUpdated Path :\n");
    for (int i = 0; i < n; i++){
        for(int j = 0; j < n; j++)
            if (d[i][j] == INFI) printf("INF\t");
            else printf("%d\t", d[i][j]);
        printf("\n");
    }
}

void main(){
    int path[N][N];
    int n;
    printf("Enter number of nodes:");
    scanf("%d",&n);
    printf("Enter distance matrix\n(enter -1 for infinity):\n");
    for (int i=0;i<n;i++)
        for(int
            j=0;j<n;j++){ int
            a; scanf("%d",&a);
            if(a==-1) path[i][j] = INFI;
```



```

        else path[i][j]= a;
    }
    floyd(path, n);
}

```

OUTPUT :

```

Enter number of nodes:4
Enter distance matrix
(enter -1 for infinity):
0 -1 3 -1
2 0 -1 -1
-1 7 0 -1
6 -1 -1 0

Updated Path :
0  10  3  INF
2  0  5  INF
9  7  0  INF
6  16  9  0

```

LEETCODE 3243

The screenshot shows the LeetCode submission page for problem 3243. The submission is accepted, with a runtime of 82ms and 52.63% of submissions being faster. The code is written in C++ and includes a macro for MAXN. The code defines a function 'shortestDistanceAfterQueries' that takes an integer 'n', a vector of queries, and returns a vector of integers. The function uses a BFS-like approach to calculate the shortest path after a series of queries.

```

/**

```

* Note: The returned array must be malloced, assume caller calls free().

```

*/

```

```

#define MAXN 505

```

```

int adj[MAXN][MAXN];

```

```

int adjSize[MAXN];

int bfs(int n) {
    int dist[MAXN];
    for (int i = 0; i < n; i++) dist[i] = -1;

    int queue[MAXN];
    int front = 0, back = 0;

    queue[back++] = 0;
    dist[0] = 0;

    while (front < back) {
        int u = queue[front++];
        for (int i = 0; i < adjSize[u]; i++)
            { int v = adj[u][i];
              if (dist[v] == -1)
                  { dist[v] = dist[u] +
                    1; queue[back++] =
                      v;
                    }
              }
    }

    return dist[n - 1];
}

int* shortestDistanceAfterQueries(int n, int** queries, int queriesSize, int* queriesColSize,
int* returnSize) {
    for (int i = 0; i < n; i++) adjSize[i] = 0;

    for (int i = 0; i < n - 1; i++)
        adj[i][adjSize[i]++] = i + 1;

    int* result = (int*)malloc(sizeof(int) * queriesSize);

```

```
for (int i = 0; i < queriesSize; i++)
    { int u = queries[i][0];
      int v = queries[i][1];

      adj[u][adjSize[u]++] = v;
      result[i] = bfs(n);
    }

*returnSize = queriesSize;
return result;

}
```

Lab program 8a :

Prim's Algorithm

```
#include <stdio.h>
#define MAX 100
#define INF 999999

void prims(int cost[MAX][MAX], int n){
    int visited[MAX] = {0}, minCost = 0, edges = 0;
    visited[0] = 1;

    while (edges < n -
        1){ int min = INF, u,
            v;
            for (int i = 0; i < n; i++)
                if (visited[i])
                    for (int j = 0; j < n; j++)
                        if (!visited[j] && cost[i][j] <
                            min){ min = cost[i][j]; u = i; v = j;
                        }

            printf("%d - %d : %d\n", u, v, min);
            minCost += min;
            visited[v] = 1;
            edges++;
        }
    printf("Minimum Cost = %d\n", minCost);
}

void main(){
    int n, cost[MAX][MAX];
    printf("Enter the number of vertices: ");
    scanf("%d", &n);
    printf("Enter the adjacency matrix:\n");
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++){
```

```

scanf("%d", &cost[i][j]);
if (i != j && cost[i][j] == 0) cost[i][j] = INF;
}

prims(cost, n);
}

```

OUTPUT:

```

Enter the number of vertices: 4
Enter the adjacency matrix:
0 10 6 5
10 0 0 15
6 0 0 4
5 15 4 0
0 - 3 : 5
3 - 2 : 4
0 - 1 : 10
Minimum Cost = 19

```

Lab program 8b:

Kruskal's Algorithm

```

#include <stdio.h>
#define MAX 100
#define INF 999999

```

```

int parent[MAX];

```

```

int find(int
    v){ while(parent[v]
    )
        v = parent[v];
    return v;
}

```

```

void kruskal(int cost[MAX][MAX], int
    n){ int edges = 0, minCost = 0;

```

```

while(edges < n - 1){
    int min = INF, u = -1, v = -1;

    for(int i = 0; i < n; i++)
        for(int j = 0; j < n; j++)
            if(cost[i][j] < min){
                min = cost[i][j];
                u = i;
                v = j;
            }

    int ru = find(u), rv = find(v);

    if(ru !=
        rv){ parent[rv] =
            ru;
            printf("%d - %d : %d\n", u, v, min);
            minCost += min;
            edges++;
        }

    cost[u][v] = cost[v][u] = INF;
}

printf("Minimum Cost = %d\n", minCost);
}

int main(){
    int n, cost[MAX][MAX];

    printf("Enter the number of vertices: ");
    scanf("%d", &n);

    printf("Enter the adjacency matrix:\n");

```

```

for(int i = 0; i < n; i++)
    for(int j = 0; j < n; j++){
        scanf("%d", &cost[i][j]);
        if(i == j || cost[i][j] == 0)
            cost[i][j] = INF;
    }

kruskal(cost, n);
return 0;
}

```

OUTPUT:

```

Enter the number of vertices: 4
Enter the adjacency matrix:
0 10 6 5
10 0 0 15
6 0 0 4
5 15 4 0
2 - 3 : 4
0 - 3 : 5
0 - 2 : 6
Minimum Cost = 15

```

Lab program 9:

Fractional Knapsack

```
#include <stdio.h>
```

```
#define MAX 15
```

```
struct
```

```
    item{ char
```

```
        name; int
```

```
        profit; int
```

```
        wt; float
```

```
        ratio;
```

```
    }a[MAX];
```

```
void swap(struct item *a, int i, int
```

```
    j){ int t[2];char ch;float f;
```

```
    t[0]=a[i].profit;
```

```
    t[1]=a[i].wt;
```

```
    ch = a[i].name;
```

```
    f=a[i].ratio;
```

```
    a[i].profit = a[j].profit;
```

```
    a[i].wt = a[j].wt;
```

```
    a[i].ratio = a[j].ratio;
```

```
    a[i].name = a[j].name;
```

```
    a[j].name = ch;
```

```
    a[j].profit = t[0];
```

```
    a[j].wt = t[1];
```

```
    a[j].ratio = f;
```

```
}
```

```
int sort(struct item* a,int st,int
```

```
    end){ float p=a[st].ratio;
```

```
    int i=st,j=end;
```

```
    while (1) {
```



```

        while (i <= end && a[i].ratio >= p) i++;
        while (j >= st && a[j].ratio < p) j--;
        if (i >= j) break;
        swap(a, i, j);
    }
    swap(a,st,j);
    return j;
}

```

```

void quickSort(struct item* a,int st,int
    end ){ if(st >= end) return;

    int mid = sort(a,st,end);
    quickSort(a,st,mid-1);
    quickSort(a,mid+1,end);
}

```

```

void fillUp(struct item* a, int cap, int
    n){ quickSort(a,0, n-1);
    float pro=0.0;
    for(int i=0;i<n && cap>0 ;
        i++){ if(cap- a[i].wt >= 0){
            cap-=a[i].wt;
            pro+= a[i].profit;
        }
        else {
            pro += cap * a[i].ratio;
            cap=0;
        }
        printf("%c ",a[i].name);
    }
    printf("\nProfit = %.2f", pro);
}

```

```

int
main(){ printf("Enter size:"); int n;
scanf("%d",&n);

printf("Enter wt, profit and name resp :\n");
for(int i=0;i<n;i++){
    scanf("%d",&a[i].wt);
    scanf("%d",&a[i].profit);
    scanf(" %c",&a[i].name);
    a[i].ratio = (float)a[i].profit / a[i].wt;
}
int cap;
printf("\nEnter capacity :");
scanf("%d",&cap);
fillUp(a, cap, n);
return 0;
}

```

OUTPUT:

```

Enter size:3
Enter wt, profit and name resp :
10 60 a
20 100 b
30 120 c

Enter capacity :50
a ,b ,c ,
Profit = 240.00

```

LEETCODE

455. Assign Cookies Solved

Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.

Each child i has a greed factor $g[i]$, which is the minimum size of a cookie that the child will be content with; and each cookie j has a size $s[j]$. If $s[j] \geq g[i]$, we can assign the cookie j to the child i , and the child i will be content. Your goal is to maximize the number of your content children and output the maximum number.

Example 1:

Input: $g = [1,2,3]$, $s = [1,1]$
Output: 1
Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3. And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed factor is 1 content. You need to output 1.

Runtime: 3 ms | Beats: 99.41%
Memory: 12.61 MB | Beats: 38.12%

```
int compare(const void *a, const void *b)
{
    return (*(int*)a - *(int*)b);
}

int findContentChildren(int* g, int gSize, int* s, int sSize)
{
    qsort(g, gSize, sizeof(int), compare);
    qsort(s, sSize, sizeof(int), compare);
    int i = 0;
    int j = 0;
    while (i < gSize && j < sSize)
    {
        if (s[j] >= g[i]) i++;
        j++;
    }
    return i;
}
```

Lab program 10:

Dijkstra's Algorithm

```
#include <stdio.h>
```

```
#define V 10;
```

```
int minDist(int visited[], int dist[], int  
n){ int min = 10000, minIdx = -1;  
for(int v=0;v<n;v++) if(!visited[v]  
    && dist[v]<min){  
    min= dist[v]; minIdx = v;  
    }  
return minIdx;  
}
```

```
void Dijkstra(int n, int graph[n][n],int  
src){ int dist[n];  
int visited[n];  
for(int  
    i=0;i<n;i++){ dist  
    [i]= 10000;  
    visited[i]=0;  
    }
```

```
dist[src] = 0;
```

```
for(int i=1;i<n;i++){  
    int u= minDist(visited, dist , n);  
    visited[u] = 1;
```

```
    for(int v=0;v<n;v++){  
        if(!visited[v] && graph[u][v] && graph[u][v]+ dist[u] < dist[v]) dist[v]  
            = dist[u] + graph[u][v];  
    }  
}
```

```
printf("Vertex\tDistance from Source :\n");
```

```

        for(int i=0;i<n;i++) printf("%d\t%d\n", i, dist[i]);

    }

```

```

void
main(){ int
    n;
    printf("Enter number of vertex:");
    scanf("%d", &n);

    int graph[n][n];
    printf("Enter weights:\n");
    for(int i=0;i<n;i++)
        for(int j=0;j<n;j++)
            scanf("%d", &graph[i][j]);
    // int graph[n][n] = {
    //     {0, 10, 0, 30, 100},
    //     {10, 0, 50, 0, 0},
    //     {0, 50, 0, 20, 10},
    //     {30, 0, 20, 0, 60},
    //     {100, 0, 10, 60, 0}
    // };
    Dijkstra(n, graph, 0);
}

```

OUTPUT:

```

Enter number of vertex:5
Enter weights:
0 10 0 30 100
10 0 50 0 0
0 50 0 20 10
30 0 20 0 60
100 0 10 60 0
Vertex Distance from Source :
0    0
1   10
2   50
3   30
4   60

```

Lab Program 11:

N Queens Problem

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int x[20];
```

```
int sol = 0;
```

```
int place(int k, int i){
```

```
    for (int j = 1; j < k; j++)
```

```
        if (x[j] == i || abs(j - k) == abs(x[j] - i))
```

```
            return 0;
```

```
    return 1;
```

```
}
```

```
void NQueens(int k, int n){
```

```
    for (int i = 1; i <= n; i++)
```

```
        {
```

```
            if (place(k, i)){
```

```
                x[k] = i;
```

```
                if (k ==
```

```
                    n){ sol++;
```

```
                    printf("\nSolution %d:\n", sol);
```

```
                    for (int row = 1; row <= n; row++)
```

```
                        { for (int col = 1; col <= n; col++){
```

```
                            if (x[row] == col)
```

```
                                printf("1 ");
```

```
                            else
```

```
                                printf("0 ");
```

```
                        }
```

```
                        printf("\n");
```

```
                    }
```

```
                }
```

```
            else
```

```

        NQueens(k + 1, n);
    }
}

int
main(){
    int n;
    printf("Enter number of queens: ");
    scanf("%d", &n);
    NQueens(1, n);
    printf("\nTotal solutions = %d\n", sol);
    return 0;
}

```

OUTPUT:

```

Enter number of queens: 4

Solution 1:
0 1 0 0
0 0 0 1
1 0 0 0
0 0 1 0

Solution 2:
0 0 1 0
1 0 0 0
0 0 0 1
0 1 0 0

Total solutions = 2

```